TEXORA · COURSE NOTES · LECTURE SHEET

Viscosity

The internal friction of fluids — Newton's law of viscous flow, Stokes' law and terminal velocity, Poiseuille flow through tubes, and why viscosity runs the print shop and the spinning hall.

PHY-1103 · PHYSICS-I · TOPIC 5 · LEVEL 1 TERM 1

1. Newton's law of viscous flow

When a fluid flows, its layers slide over one another at different speeds, and friction between the layers resists the sliding — that internal friction is **viscosity**. Newton proposed, and experiment confirms for most ordinary fluids (the **Newtonian** fluids), that the frictional force between layers is proportional to their area and to the **velocity gradient** — the rate at which speed changes perpendicular to the flow:

$$F = \eta A (dv/dx), \quad \text{equivalently } \eta = (F/A) / (dv/dx) = \text{shear stress / velocity gradient}$$

The proportionality constant η is the **coefficient of viscosity**, with SI unit Pa s (the older unit, the poise, is 0.1 Pa s). Scale anchors: water ≈ 1.0×10^{-3} Pa s, olive oil ≈ 0.1 Pa s, glycerine ≈ 1 Pa s, air ≈ 1.8×10^{-5} Pa s — five orders of magnitude across everyday fluids. Some materials refuse Newton's straight line altogether: printing pastes, polymer melts and many textile auxiliaries are **non-Newtonian**, thinning under shear (pseudoplastic — they flow when the squeegee pushes and stand stiff when it rests) or thickening; their "viscosity" is a curve, not a number, and §6 shows why the industry cares.

2. Stokes' law and terminal velocity

A small sphere moving slowly through a viscous fluid feels a drag that Stokes computed exactly:

$$F = 6\pi \eta r v$$

A falling sphere accelerates until the drag plus buoyancy balance its weight; thereafter it falls at the constant **terminal velocity**. Balancing the three forces — weight $(4/3)\pi r^3 \rho g$, upthrust $(4/3)\pi r^3 \sigma g$ (σ = fluid density, Topic 4), and Stokes drag — gives

$$v_t = 2 r^2 (\rho - \sigma) g / (9 \eta)$$

WORKED EXAMPLE — A STEEL BALL IN OIL

A steel ball of radius 1.0 mm (ρ = 7800 kg m^{-3}) falls through oil (σ = 900 kg m^{-3}, η = 0.20 Pa s). $v_t = 2 \times (10^{-3})^2 \times (7800 - 900) \times 9.8 / (9 \times 0.20) = 2 \times 10^{-6} \times 6900 \times 9.8 / 1.8 = 0.135 / 1.8 \approx$ **0.075 m s^{-1}** — about 7.5 cm per second, slow enough to time with a stopwatch between two marks. Invert the arrangement — time the ball, know the densities — and Stokes' law becomes a viscometer: the falling-ball method of measuring η.

Stokes' law holds only for small spheres, slow (laminar) motion and an effectively unbounded fluid; walls and turbulence each need corrections — examination answers should say so.

3. Poiseuille's law: flow through a tube

For laminar flow of a viscous liquid through a horizontal capillary tube, Poiseuille integrated the velocity profile — parabolic, fastest at the axis, zero at the wall — to give the volume flow rate:

$$Q = \pi p r^4 / (8 \eta l)$$

where p is the pressure difference between the ends, r the tube radius and l its length. The law's famous feature is the **fourth power of the radius**: halve the radius and the flow falls sixteen-fold at the same pressure. It is the sessional course's own experiment (viscosity by capillary flow), the design rule for every tube, nozzle and pore a process liquid must pass, and the reason a partly blocked filter or needle announces itself so drastically.

WORKED EXAMPLE — POISEUILLE FLOW

Water ($\eta = 1.0 \times 10^{-3}$ Pa s) flows through a capillary of radius 0.50 mm and length 0.50 m under a pressure difference of 1000 Pa. $Q = \pi \times 1000 \times (5.0 \times 10^{-4})^4 / (8 \times 10^{-3} \times 0.50) = \pi \times 1000 \times 6.25 \times 10^{-14} / (4.0 \times 10^{-3}) \approx$ **4.9 × 10^{-8} m^3 s^{-1}** — about 0.05 mL per second. Now halve the radius: Q falls by $2^4 = 16$, to $\approx 3 \times 10^{-9}$ m^3 s^{-1}. The r^4 is the whole lesson.

4. Streamline limits, temperature, and measurement

Viscous (streamline) descriptions hold below the **critical velocity** at which turbulence sets in — the Reynolds-number borderland of Topic 4 (Re ≈ 2000 in pipes): above it, drag laws change character and Poiseuille's formula fails. Temperature moves viscosity in opposite directions for the two fluid families: for **liquids**, η falls steeply with temperature (cohesion loosens — warm oil pours, cold oil crawls, and η of water roughly halves between 0 °C and 40 °C), while for **gases** η rises with temperature (faster molecules transport more momentum between layers). Measurement rounds out the topic: the capillary (Ostwald/Poiseuille) viscometer, the falling-ball (Stokes) method, and the rotating-cylinder (Couette) instrument — three faces of the three laws of this sheet.

5. The textile thread

Viscosity is a process specification everywhere in the mill. **Printing paste** viscosity — engineered with thickeners to be pseudoplastic — decides whether a design prints with a razor edge or bleeds into the ground: shear-thinning lets the paste pass the screen, instant recovery holds the line. **Sizing** recipes are viscosity-controlled so the size film coats the warp yarn evenly (Topic 3's wetting gets it in; this topic's viscosity sets how it stays). **Melt spinning** extrudes polymer whose melt viscosity, set by temperature and chain length, determines whether filaments form or drip. Dye bath circulation, lubricant oils on spinning machinery, coating and laminating pastes — each has a viscosity window written into its process sheet, and each window is this topic's three laws at work.

6. Glossary — key terms of this topic

TERM	MEANING
Viscosity (η)	A fluid's internal friction; the ratio of shear stress to velocity gradient. Unit Pa s.
Velocity gradient	The rate of change of flow velocity perpendicular to the flow, dv/dx (s^{-1}).
Newtonian fluid	A fluid whose η is constant, independent of the shear rate (water, oils, air).
Stokes' law	Drag on a small sphere in slow viscous motion: $F = 6\pi\eta rv$.
Terminal velocity	The constant speed at which drag + upthrust balance a falling body's weight: $v_t = 2r^2(\rho - \sigma)g/9\eta$.
Poiseuille's law	Laminar flow through a tube: $Q = \pi pr^4/8\eta l$ — note the fourth power of radius.
Critical velocity	The flow speed (set by Reynolds number) above which laminar flow breaks into turbulence.
Pseudoplastic	Shear-thinning: apparent viscosity falls as shear rate rises (printing pastes, polymer melts).

7. Key points and summary

- Viscosity is inter-layer friction: $F = \eta A(dv/dx)$; η spans 10^{-5} (air) to 1+ Pa s (glycerine) and defines the Newtonian family.
- Stokes' drag $6\pi\eta rv$ gives terminal velocity $v_t = 2r^2(\rho - \sigma)g/9\eta$ — a viscometer in disguise.
- Poiseuille's $Q = \pi pr^4/8\eta l$ rules all tube flow, with its unforgiving r^4; temperature lowers liquids' viscosity and raises gases'; turbulence ends the regime where these laws hold.

8. Practice

Answer in your own words or with full working; brief model answers follow.

1. A fluid layer of area 0.10 m^2 experiences a viscous force of 0.020 N with a velocity gradient of 20 s^{-1} across it. Find η and its unit.
2. Using the worked example of §2, find the terminal velocity if the ball's radius is doubled (all else unchanged), and explain the direction of the change physically.
3. A capillary delivers 8.0×10^{-8} m^3 s^{-1} of a liquid. The tube is replaced by one of half the radius and twice the length, at the same pressure difference. Find the new flow rate.
4. Why does Poiseuille's law fail once the flow becomes turbulent?
5. Glycerine is poured on a cold morning and on a hot afternoon. Which pour is faster and why — and why does the same comparison run the other way for air in a hot pipe?
6. A printing paste must pass a screen easily under the squeegee yet hold a sharp edge at rest. Name the flow behaviour required and sketch (in words) its η-versus-shear-rate curve.
7. State the three forces balanced at terminal velocity, and the conditions under which Stokes' law setting them equal is trustworthy.
8. Machine oil in a gearbox warms up during a shift. State two consequences of the viscosity change for the machine's running.

Answer key

1. $\eta = (F/A)/(dv/dx) = (0.020/0.10)/20 = 0.20/20 = 0.010$ Pa s.
2. $v_t \propto r^2$, so doubling r quadruples v_t to ≈ 0.30 m s^{-1}. Physically, weight grows as r^3 while Stokes drag grows only as r, so a bigger sphere needs more speed before drag catches up.
3. $Q \propto r^4/l$: factor $(1/2)^4 \times (1/2) = 1/32$. New $Q = 8.0 \times 10^{-8} / 32 = 2.5 \times 10^{-9}$ m^3 s^{-1}.
4. Its derivation assumes laminar, layered flow with a parabolic profile; turbulence mixes the layers, destroys the profile, and the pressure–flow relation changes character (roughly $Q \propto \sqrt{p}$ instead).
5. The hot pour: liquid viscosity falls steeply with temperature as cohesion weakens. For gases the mechanism is molecular momentum transport, which faster (hotter) molecules do more of — gas viscosity rises with temperature.
6. Pseudoplastic (shear-thinning) behaviour: η high at rest (sharp edge holds), falling as shear rate rises under the squeegee (easy passage), recovering when shear stops.
7. Weight of the sphere, upthrust of the displaced fluid, and Stokes drag — equal when acceleration ceases. Trustworthy for small spheres, slow laminar motion, in a fluid effectively unbounded (with wall corrections otherwise).

8. The oil’s viscosity falls as it warms: frictional drag and power loss in the bearings fall (easier running), but the protective oil film also thins, so wear protection drops — the two-sided reason gearboxes specify oil grades for working temperature.
